

Mini Project 1.2: FDD versus TDD Under Traffic Asymmetry

1. Introduction and model

Section 1.8 describes FDD as two fixed, paired channels and TDD as one channel shared in time, with guard periods and synchronization as the price for a flexible uplink/downlink split. This project quantifies that trade-off with a Python simulator.

- **Capacity:** both schemes have the same total capacity of 100 normalized units. FDD gives a fixed 50 units to each direction.
- **TDD frame:** 20 slots per frame, of which g are guard slots that carry no traffic. The remaining slots are split between downlink and uplink, and the simulator picks the split that serves the most traffic, keeping at least one slot in each direction.
- **Demand:** offered load equals 100% of total capacity, split by a DL:UL ratio. Utilization is served traffic divided by total capacity. Guard overhead was tested at $g = 1, 2$ and 4 slots (5%, 10%, 20%).

2. Results

Demand (DL:UL)	FDD	TDD, 5% guard	TDD, 10% guard	TDD, 20% guard
50:50	100% (50/50)	95% (45/50)	90% (40/50)	80% (30/50)
70:30	80% (50/30)	95% (65/30)	90% (60/30)	80% (50/30)
90:10	60% (50/10)	95% (85/10)	90% (80/10)	80% (70/10)

Table 1: Utilization, with effective DL/UL throughput in brackets (normalized units).

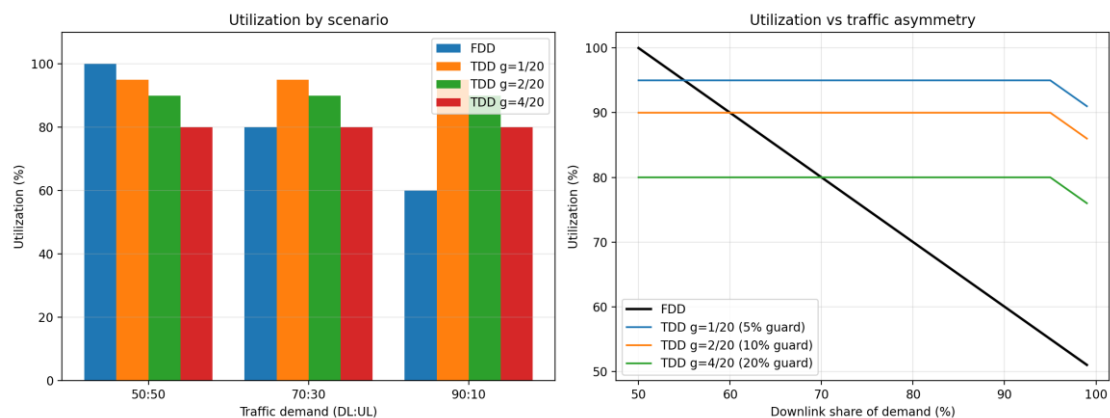


Figure 1: Left, utilization for the three required scenarios. Right, utilization across the full asymmetry sweep.

3. Analysis

FDD wastes capacity when traffic is asymmetric. Its 50-unit uplink cannot lend capacity to the downlink, so at 90:10 the uplink demand uses only 10 of its 50 units and utilization falls to 60%. Analytically, for a downlink share p above 0.5, FDD utilization is $150 - 100p$ percent, so it declines steadily as asymmetry grows.

TDD pays a fixed guard tax but tracks demand. By re-slotting to 17/2 at 90:10, TDD serves 95% of the offered load with a 5% guard. Its utilization stays near $(1 - \text{guard fraction})$ at every asymmetry, so it is flat where FDD is falling.

Crossover. TDD overtakes FDD from about 56:44 (DL:UL) with a 5% guard, about 61:39 with 10%, and about 71:29 with 20%. Larger switching overhead therefore erodes TDD's flexibility advantage, and at 20% guard TDD only ties FDD at 70:30 and loses clearly at 50:50 (80% versus 100%). FDD remains competitive or better for near-symmetric traffic, up to roughly 55:45 to 70:30 depending on the guard size.

When TDD's advantage is largest. The gap is widest at strong downlink asymmetry with small guard overhead: at 90:10 with a 5% guard, TDD delivers 95% against FDD's 60%, a 35-point gain. Since real mobile traffic is heavily downlink-dominated, this is the practically relevant region.

4. Trade-offs from Section 1.8

- **Paired spectrum:** FDD needs two separated blocks of spectrum, which are scarce. TDD needs only one, and can use unpaired bands.
- **Synchronization:** TDD requires all nearby base stations to agree on frame timing, otherwise uplink and downlink transmissions interfere with each other. FDD has no such requirement.
- **Guard periods:** they are TDD's direct capacity cost, modeled here as dead slots. They also grow with cell size, because larger cells have longer propagation delays.
- **Slot-ratio flexibility:** TDD can adapt to demand, as shown in the results. FDD's fixed split cannot. FDD also transmits continuously in both directions, which suits delay-sensitive traffic.

5. Limitations

The model assumes a saturated network (offered load equals capacity), a single fixed traffic ratio per run, and guard time as a whole-slot fraction. It ignores scheduling delay, retransmissions and the fact that real slot ratios come from a small set of standardized configurations. At lower offered loads FDD's wasted capacity would matter less, since the downlink might not saturate. The conclusions about crossover points should therefore be read as indicative, not exact.